

Secure Exit

CO3-AT3-Assignment

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SIMATS Online Programming Lab

JAVA PROGRAMMING

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QUESTIONS

Q1

Banking Transaction System –
Exception Handling

10 marks



Q2

Hospital Patient Registration –
User-Defined Exception

10 marks



Q3

E-Commerce Order Processing
– Built-in Exceptions

10 marks



Q4

Hospital Emergency System –
Thread Priority

10 marks



Q5

Online Food Delivery – Multiple
Threads and Execution Order

10 marks



5/5 answered

Q1 Banking Transaction System – Exception Handling

10 marks

Develop a Java program for an ATM withdrawal system. The program accepts the account balance and withdrawal amount.

The program must:

- Accept the current account balance and withdrawal amount.
- Throw a user-defined `InsufficientBalanceException` when the withdrawal amount exceeds the balance.
- Handle withdrawal amounts that are zero or negative.
- Handle invalid numeric input using a suitable built-in exception.
- Display a meaningful transaction result.
- Use try-catch-finally so that the transaction completion message is always displayed.

Sample Test Cases:

Input / Scenario Expected Result Purpose

10000 3000 Normal case

10000 12000 Boundary / exception case

10000 -500 Invalid input

abc 3000 Invalid input

Your Code

```
1 import java .util.*;
2 class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         try{
6             int balance=sc.nextInt();
7             int amount=sc.nextInt();
```

```
8         if (amount <= 0) {
9
10            throw new IllegalArgumentException("withdraw amount.");
11        }
12        if (amount > balance) {
13            throw new InsufficientBalanceException("Insufficient.");
14        }
15        balance = balance - amount;
16        System.out.println("Withdrawl Successful.");
17        System.out.println("Remaning Balance = " + balance);
18
19    }
20    catch (InsufficientBalanceException e) {
21        System.out.println(e.getMessage());
22    }
23    catch (IllegalArgumentException e) {
24        System.out.println(e.getMessage());
25    }
26    catch (Exception e) {
27        System.out.println("Invalid input.");
28    }
29    finally {
30        System.out.println("Transaction complete.");
31    }
32 }
33 }
34 class InsufficientBalanceException extends Exception {
35     InsufficientBalanceException(String message) {
36         super(message);
37     }
38 }
```

Input

abc 3000



Output

Invalid input.
Transaction complete.

✔ Submitted

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